Mehraan Chowdhury

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SUMMARY

- Master of Philosophy graduate from UWA with thesis on deep learning for medical screening, supported by 2 first-author IEEE conference publications
- Master of Information Technology (Artificial Intelligence) from University of Melbourne, WAM 81.1 (H1 / High Distinction average)
- 3.5 years of commercial front-end development experience building responsive cross-browser interfaces in HTML5, CSS3, JavaScript, and jQuery
- Proficient in Python, Java, C, and SQL with hands-on experience deploying deep learning pipelines using PyTorch, scikit-learn, and OpenCV
- Effective written and verbal communicator with experience presenting technical research at international IEEE conferences in the USA and New Zealand

EDUCATION

Master of Philosophy (Computer Science) Sep 2023 - Mar 2026
University of Western Australia

- Thesis: *Deep Learning-Based Screening of Obstructive Sleep Apnea Using 3D Facial Images*
- Ranked in the top 5% of theses examined, as noted in the examiner report
- Two first-author IEEE conference publications produced from thesis research

Master of Information Technology (Artificial Intelligence) Mar 2021 - Dec 2022
University of Melbourne

- WAM: 81.1 (H1 / High Distinction average)
- Thesis: *Predicting Relevant Context for Programming Questions using Stack Overflow Answers*

Bachelor of Science (Neuroscience)
University of Melbourne, Mar 2013 - Nov 2015

PUBLICATIONS

- M. Chowdhury, M. Bennamoun, F. Boussaid, S. M. Shamsul Islam, "Classifying Obstructive Sleep Apnea from 3D Facial Images Using Automated Landmark Detection and Geodesic Distances," *2025 IEEE Conference on Future Machine Learning and Data Science (FMLDS)*, Los Angeles, 2025, pp. 602-607. doi: 10.1109/FMLDS67896.2025.00098
- M. Chowdhury, M. Bennamoun, F. Boussaid, S. M. Shamsul Islam, "Automatic Detection of Peripheral Facial Landmarks Using 3D Facial Data," *2024 IVCNZ Conference on Image and Vision Computing New Zealand*, Christchurch, 2024, pp. 1-6. doi: 10.1109/IVCNZ64857.2024.10794203

PROFESSIONAL EXPERIENCE

Applied Machine Learning Researcher Sep 2023 - Mar 2026
University of Western Australia, Perth

- Reviewed the literature for deep learning techniques for classifying 3D meshes and point clouds, identifying high-performing methods that had not been applied to this problem, and designed a program of experiments to test whether they improved classification performance
- Developed an annotation process enabling medical experts with no technical background to label 3D scans, producing the ground-truth dataset; validated the resulting annotations and prepared the data for training
- Adapted two open-source deep learning codebases, a graph convolutional network and a multi-view CNN, to a new landmarking task, training both models from scratch to locate anatomically significant points on 3D faces
- Developed an end-to-end pipeline to screen for obstructive sleep apnea from 1,309 3D facial scans, extracting 6,624 geometric features via mesh graph traversal and benchmarking 16 machine learning algorithms

under stratified 10-fold cross-validation

- Fine-tuned a generative pre-trained point cloud transformer (PointGPT) for classification and built a multi-view CNN combining 3D geometry with texture, improving AUC from 0.77 to 0.81 over the geometric feature baseline
- Built reproducible workflows on GPU hardware and authored 2 first-author IEEE conference papers from the resulting research
- **Tools:** Python, PyTorch, scikit-learn, NumPy, Optuna, Open3D, trimesh, MeshLab, CloudCompare, NetworkX, Linux, Git

Graduate Researcher (Natural Language Processing) 2022

University of Melbourne, Melbourne

- Built an NLP pipeline in Python to predict relevant programming context from Stack Overflow answers using transformer models (Hugging Face) and NLTK
- Scraped and processed large volumes of developer Q&A data using Beautiful Soup and ChromeDriver
- Reported findings in a thesis contributing to a final WAM of 81.1

Front-End Web Developer Apr 2016 - Dec 2019

Revo Interactive Ltd, Dhaka, Bangladesh

- Converted design wireframes into responsive production user interfaces using HTML5, CSS3, JavaScript, and jQuery
- Ensured cross-browser consistency across Chrome and Firefox
- Integrated SEO content into HTML markup to improve search visibility for client websites
- Diagnosed and resolved bugs across the front-end codebase, reducing time-to-fix on reported issues
- Contributed design and development ideas during sprint planning, helping shape feature direction across the team

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, HTML5, CSS3, SQL, Bash
- **ML & Data:** PyTorch, scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Optuna, Hugging Face, NLTK, Jupyter
- **Computer Vision & 3D:** OpenCV, PIL, Open3D, trimesh, MeshLab, CloudCompare
- **Web automation:** Beautiful Soup, Selenium / ChromeDriver, Playwright
- **Tools & Methods:** Git, GitHub, Linux, LaTeX, Agile, Waterfall

REFERENCES

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